

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch : IT Semester : VII Subject Code & Name: Real Time System [BTITPE705A] Max Marks: 60 Date:9/02/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain Soft real system and Hard real system.		6
B)	Explain reference model of real system.		6
C)	Explain real time system scheduling with its types.		6
Q.2	Solve Any Two of the following.		12
A)	What is clock driven scheduling? Explain static clock driven scheduling.		6
B)	Explain the general structure of cyclic schedule.		6
C)	How we improve average response time in clock driven scheduling? Explain it.		6
Q. 3	Solve Any Two of the following.		12
A)	Write the difference between Fixed priority and Dynamic priority driven scheduling.		6
B)	Explain the optimality of the RM and DM algorithms.		6
C)	What is schedulability test for fixed-priority tasks with short response times? Explain.		6
Q.4	Solve Any Two of the following.		12
A)	Write the approaches of periodic and sporadic jobs in priority-driven systems.		6
B)	Write a note on sporadic servers.		6
C)	Explain resource access control and resource contention in details.		6
Q. 5	Solve Any Two of the following.		12
A)	Explain Basic Priority-Inheritance protocol in details.		6
B)	Explain model of multiprocessor and distributed system.		6
C)	Write about priority-ceiling protocol.		6
*** End ***			